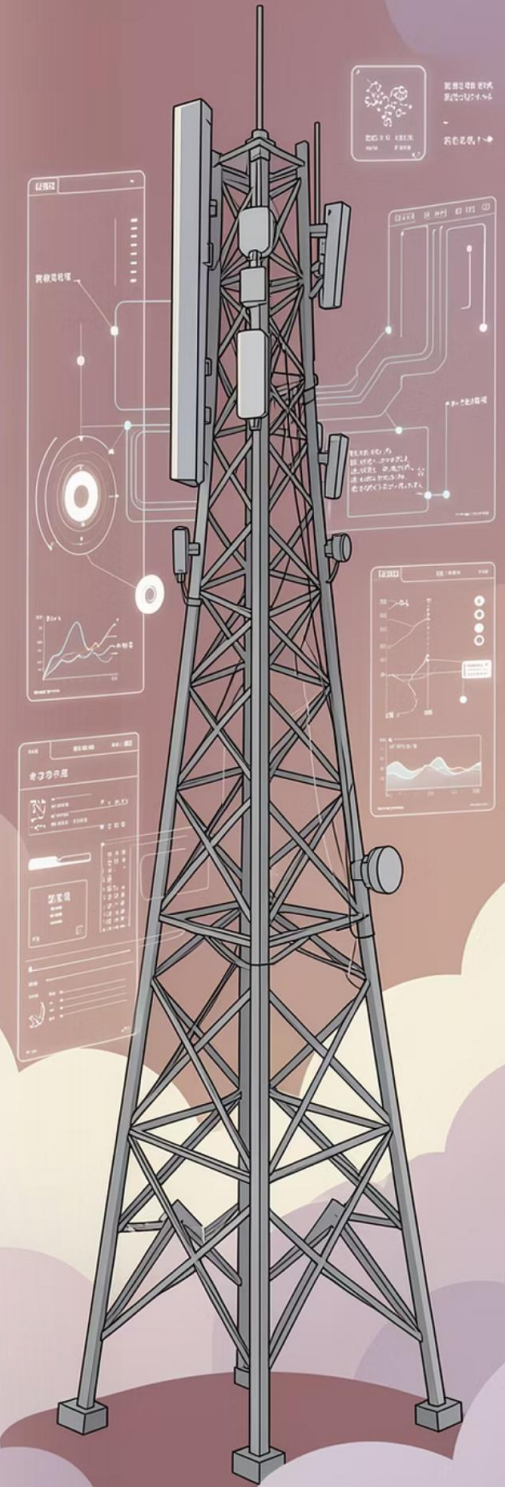


OutageIntel AI: Telecom Incident Knowledge Hub

Semantic Retrieval for Telecom Incident Intelligence



The Incident Intelligence Challenge



Information Overload

Engineers struggle to locate relevant past outage incidents across scattered documentation and logs



Keyword Search Limitations

Traditional keyword-based systems fail to understand context in complex telecom log data



Fragmented Knowledge

Critical incident knowledge spans PDFs, simulation logs, runbooks, and RCA documents



Urgent Response Needs

Troubleshooting demands fast, accurate, context-aware retrieval to minimize downtime

Strategic Objectives

01

Semantic Search Infrastructure

Build intelligent search over comprehensive outage documentation

02

Similar Incident Retrieval

Retrieve contextually similar past incidents using vector embeddings

03

Context-Aware Generation

Generate accurate answers using large language models

04

Operational Efficiency

Improve troubleshooting speed and decision-making accuracy



System Architecture Overview

End-to-end pipeline powered by Retrieval-Augmented Generation (RAG)



Data Ingestion

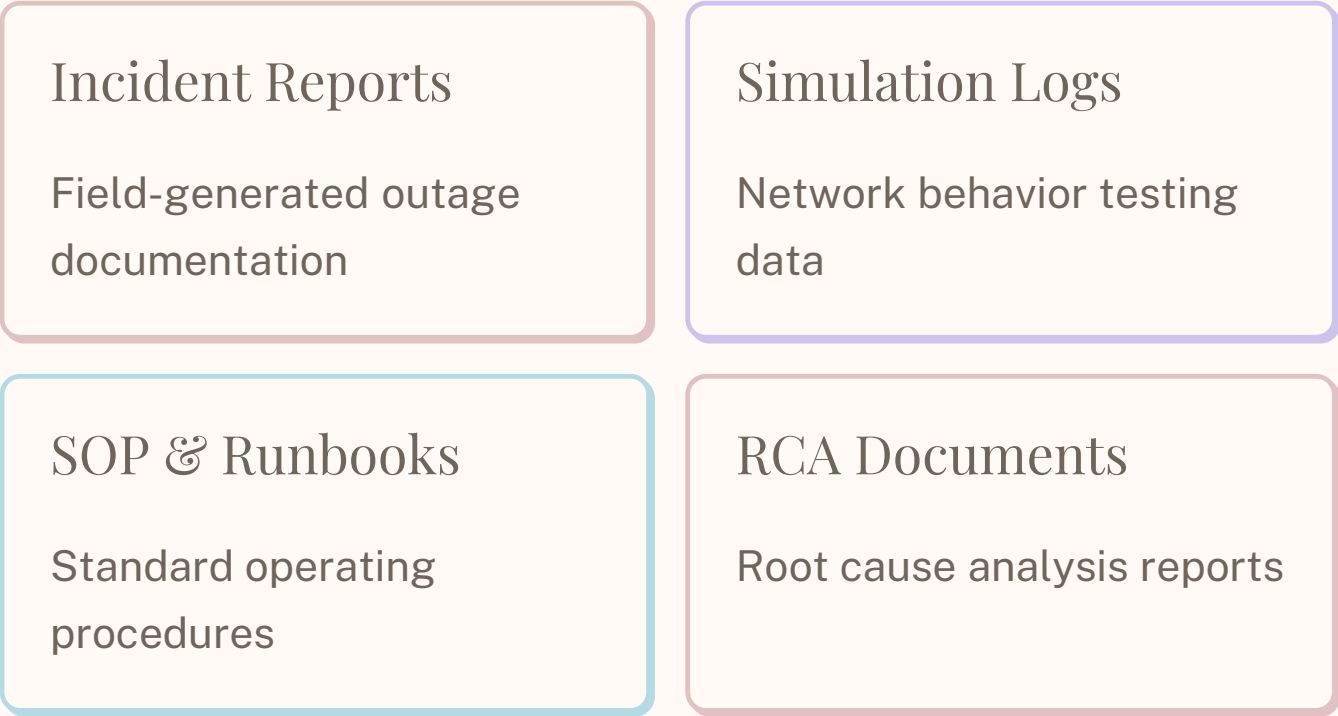
Document Processing

Embedding Generation

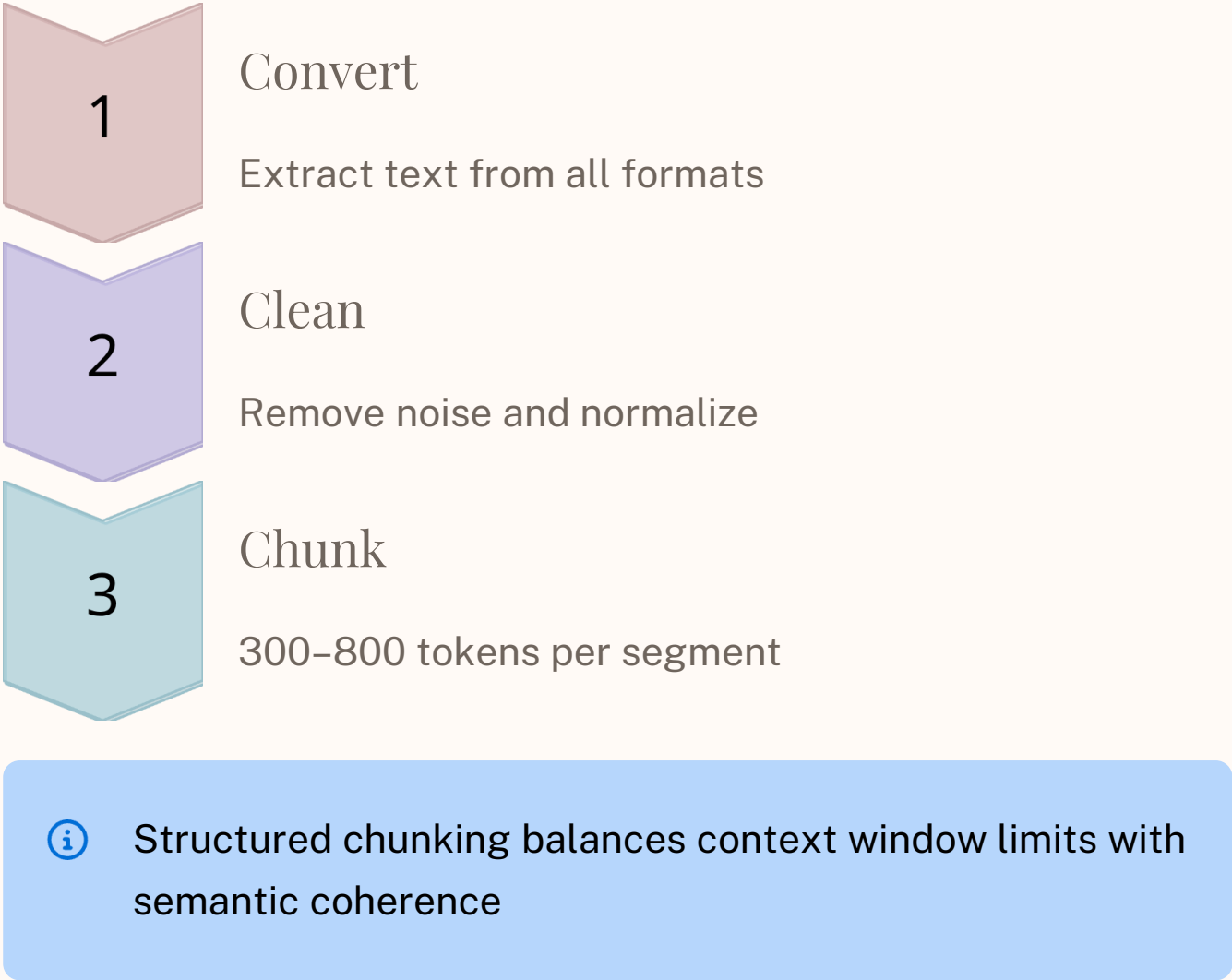
Vector DB Storage

Data Ingestion Layer

Document Sources



Processing Pipeline



Embedding Generation Layer

Transforms text into dense vector representations capturing semantic meaning

OpenAI Embeddings

Industry-standard models for semantic understanding

Sentence Transformers

Open-source alternatives with fine-tuning capabilities

Each document chunk becomes a high-dimensional vector enabling meaning-based search



Vector Database Storage

Storage Solutions

FAISS

Facebook AI Similarity Search for efficient vector indexing

Pinecone

Managed vector database with production-ready API

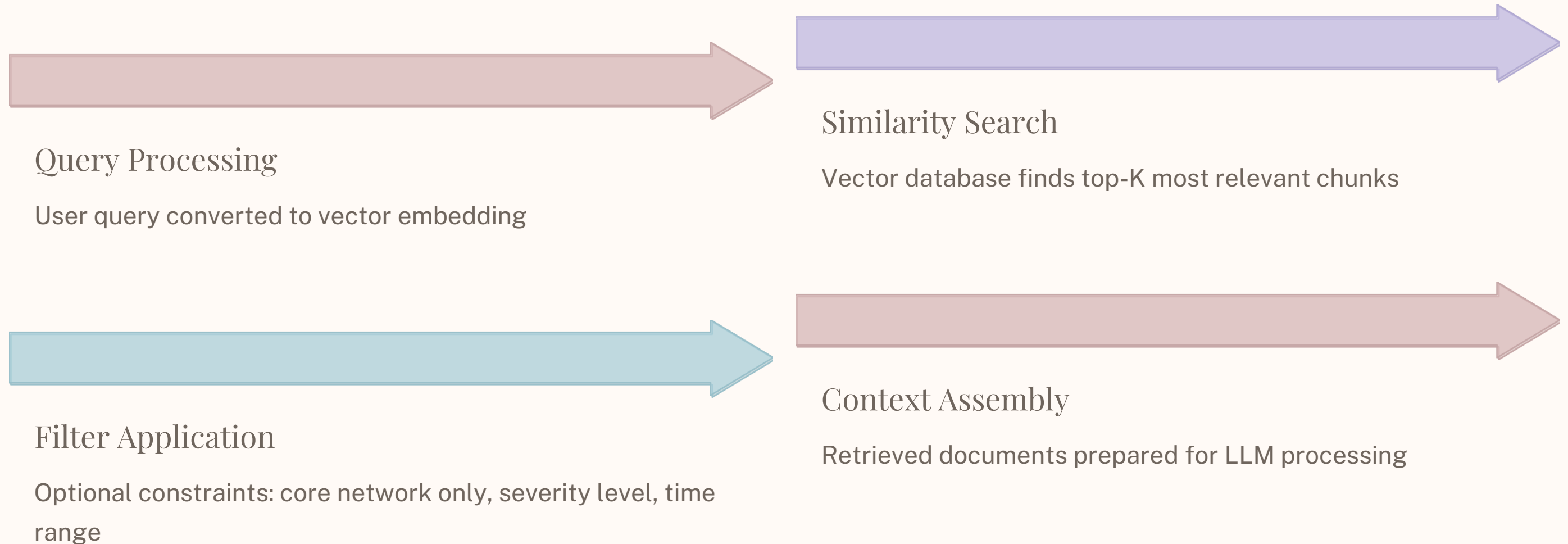
Weaviate

Open-source vector search with GraphQL interface

Metadata Fields

- Incident type classification (BGP failure, fiber cut, power loss)
- Geographic region and network topology
- Severity level and impact assessment
- Timestamp and duration tracking
- Affected equipment and services

Intelligent Retrieval Layer



📄 Example query: "BGP route flapping simulation in core routers"

LLM Generation Layer

Language Models

GPT-4

OpenAI's most capable
multimodal model

Llama 3

Open-source alternative
with commercial licensing

Prompt Engineering

Role: Telecom operations assistant

Context: Retrieved incident documents

Constraint: Answer using only provided context

Output: Concise troubleshooting guidance



Key Features

- Uses only retrieved context for accuracy
- Grounded responses prevent hallucinations
- Context-aware troubleshooting guidance
- Consistent response format



Business Impact & Benefits



Accelerated Analysis

Reduce incident analysis time from hours to minutes with intelligent document retrieval



Downtime Reduction

Minimize network outages through faster root cause identification and resolution



Knowledge Sharing

Enable cross-team collaboration and expertise transfer across regional operations



Scalable Intelligence

Build foundation for autonomous network operations and self-healing systems